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# PERFORMANCE SPECIFICATION

# FAN, VENTILATING, VEHICULAR COMPARTMENT

This specification is approved for use by the U.S. Army Tank Automotive Research, Development and Engineering Center (TARDEC), Research, Development and Engineering Command (RDECOM), Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

## 1. SCOPE

1.1 Scope. This specification covers an electric-powered, single-stage, vaneaxial fan, referred to herein as "fan", used to ventilate the crew compartment of an armored military vehicle (see 6.1).

## 2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3, 4, or 5 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents cited in sections 3, 4, or 5 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract (see 6.2).

Comments, suggestions, or questions on this document should be addressed to U.S. Army RDECOM, Tank Automotive Research, Development and Engineering Center, ATTN: RDTA-EN/STND/TRANS MS #268, 6501 E. 11 Mile Road, Warren, MI 48397-5000 or emailed to DAMI_STANDARDIZATION@conus.army.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.daps.dla.mil.

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FEDERAL STANDARDS

FED-STD-595/24533 - Green

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-461 - Requirements for the Control of Electromagnetic Interference Characteristics of Subsystems and Equipment

(Copies of these documents are available from https://assist.daps.dla.mil/quicksearch/ or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract (see 6.2).

U.S. ARMY TACOM DRAWINGS

12292099 - Fan, Ventilating

(Copies of these documents are available from DAMI_STANDARDIZATION@conus.army.mil or U.S. Army RDECOM, Tank Automotive Research, Development and Engineering Center, ATTN: RDTA-EN/STND/TRANS MS #268, 6501 E. 11 Mile Road, Warren, MI 48397-5000.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract (see 6.2).

AIR MOVEMENT AND CONTROL ASSOCIATION (AMCA)

AMCA-STD-210 - Laboratory Methods of Testing Fans for Rating

(Copies of this document are available from www.amca.org or Air Movement and Control Association, 30 West University Drive, Arlington Heights, IL 60004.)

AMERICAN SOCIETY FOR QUALITY (ASQ)

ANSI/ASQ Z1.4 - Sampling Procedures and Tables for Inspections by Attributes

(Copies of this document are available from www.asq.org or American Society for Quality, 600 North Plankinton Avenue, Milwaukee, WI 53203.)



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ASTM INTERNATIONAL

| | | |
|---|---|---|
| ASTM D149 | - | Standard Test Method for Dielectric Breakdown Voltage and Dielectric Strength of Solid Electrical Insulating Materials at Commercial Power Frequencies |
| ASTM D257 | - | Standard Test Methods for DC Resistance or Conductance of Insulating Materials |

(Copies of these documents are available from www.astm.org or ASTM International, P.O. Box C700, West Conshohocken, PA 19428-2959.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

## 3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a first article sample shall be subjected to first article inspection in accordance with 4.3.

3.2 Materials. Materials used shall be in accordance with the manufacturer's materials specifications for fans. The materials shall be capable of meeting all the operational and environmental requirements specified herein (see 4.8.1).

3.2.1 Recycled, recovered, or environmentally preferable materials. Recycled, recovered, or environmentally preferable materials should be used to the maximum extent possible provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.3 Design and construction. The fan shall be constructed in accordance with Drawing 12292099 and as specified herein (see 4.8.1 and 4.8.2).

3.3.1 Weight. The weight of the fan shall not exceed 12 pounds (lb) (see 4.8.3).

3.3.2 Electric motor. The motor shall be a series-wound, three-speed, variable, 24 volt (V) direct current (dc) type, directly coupled to the fan shaft. The motor shall be operable from a 28±5 V supply (see 4.8.1 and 4.8.2).

3.4 Performance.

3.4.1 Voltage. The fan shall operate with any control voltage from 11 to 30 V dc (see 4.8.4.1).



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3.4.2 Operational performance. The fan shall show no evidence of an abnormal noise or vibration when used within the standard operating range. The direction of airflow shall be as specified on Drawing 12292099. The current shall not exceed 29.5 amperes (A) (see 4.8.4.2).

3.4.3 Rated performance. The fan shall deliver air at the rate specified in table I under the conditions specified therein, at 29.5 A or less (see 4.8.4.3).

TABLE I. Rated performance.

| Minimum airflow rate | | Conditions | | |
|---|---|---|---|---|
| cubic meters per minute | cubic feet per minute | Input voltage (V dc) 1/ | Minimum static pressure | |
| | | | Pa | inches of water (gage) |
| 21.8 | (770) | 27 | 672 | (2.7) |
| 18.0 | (635) | 27 | 1,045 | (4.2) |
| 17.0 | (600) | 19.8 | 473 | (1.9) |
| 9.9 | (350) | 11.8 | 274 | (1.1) |

1/ Tolerance: +0.5 V dc, -0.0 V dc.

3.4.4 Electromagnetic interference suppression. The fan shall meet the requirements of MIL-STD-461 limits CE102 and RE102-3 (see 4.8.4.4).

3.4.5 Insulation resistance. The insulation resistance of the fan shall be not less than 12 megohms at 500 V dc at a relative humidity of 40 to 80 percent (%) (see 4.8.4.5).

3.4.6 Dielectric strength. The fan shall withstand a voltage of 500 V root-mean-square (rms) at 60 Hertz (Hz) for 1 minute with no evidence of arcing or breakdown (see 4.8.4.6).

3.4.7 Noise level. The sound power level of the fan, when operating as specified in 3.4.3, shall not exceed the values shown in table II, when measured at a distance of 3.048 meters (10 feet) in any direction (see 4.8.4.7).

3.4.8 Environmental.

3.4.8.1 High temperature. The fan shall demonstrate no performance degradation and show no evidence of damage or deformation after being stored at temperatures up to 71 degrees Celsius (°C) (160 degrees Fahrenheit (°F)) and operated at temperatures up to 54 °C (130 °F) (see 4.8.4.8.2).

3.4.8.2 Low temperature. The fan shall demonstrate no performance degradation and show no evidence of damage or deformation after being stored at temperatures down to minus (-) 57 °C (-70 °F) and operated at temperatures down to -54 °C (-65 °F) (see 4.8.4.8.3).



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TABLE II. Sound power level.

| Octave band (Hz) | Sound power level -12 (dB re 10 W) 1/ |
|---|---|
| 63 | 84 |
| 125 | 97 |
| 250 | 88 |
| 500 | 99 |
| 1000 | 96 |
| 2000 | 97 |
| 4000 | 92 |

1/ Reference (re), watt (W)

3.4.8.3 Steam and waterjet cleaning. The fan shall show no evidence of damage or deterioration and demonstrate no performance degradation following a steam and waterjet cleaning process which uses a cleaner conforming to best commercial practice, and which applies the jet perpendicular to the fan surface from a distance of not more than 30.5 centimeters (cm) (1 foot) for steam cleaning and not more than 1.52 meters (m) (5 feet) for waterjet cleaning. The jet pressure shall be 724±34 kPa (105±5 pounds per square inch gage (psig)) and the jet shall be moved at such a rate that the entire surface of the fan is subjected to the jet for a period of 1 minute (see 4.8.4.8.4).

3.4.8.4 Humidity. The fan shall demonstrate no performance degradation and show no evidence of damage or deformation after exposure to a relative humidity ranging from 0 to 98% (see 4.8.4.8.5).

3.4.8.5 Shock. The fan shall show no evidence of damage or deformation and demonstrate no performance degradation after exposure to sawtooth shock pulses of 40 gravity units (g) for 11 milliseconds (ms) time duration applied along each of the three major axes (see .8.4.8.6).

3.4.8.6 Vibration. The fan shall show no evidence of damage or deformation and demonstrate no performance degradation during and after exposure to vibration levels of 1.5 and 4.2 g in the frequency range of 5 to 500 to 5 Hz with a sweep time of 15 minutes (see figure 1 and 4.8.4.8.7).

3.4.9 Durability. The fan shall be capable of operating 1000 hours continuously at 27±2 V dc and 498±50 Pa (2±0.2 inches of water) pressure without performance degradation and without any maintenance or malfunction (except brushes may be replaced after not less than 500 hours). During the 1000 hours, the fan shall be in an ambient temperature of 21±11 °C (70±20 °F) and shall be switched off and on every 24 hours. A 1 minute off time shall be employed (see 4.8.4.9).

3.5 Finish. Exposed ferrous and aluminum surfaces shall be cleaned, treated, primed and finished in accordance with best commercial practice (see 4.8.2). The finish color shall match color chip number 24533 (green) of FED-STD-595.



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3.6 Identification marking. Nameplates and product marking shall be as specified on Drawing 12292099 (see 4.8.2).

3.7 Workmanship. The fan shall be free of defects such as burrs, sharp edges, corrosion, soil, dirt and any defects that would affect its performance or appearance (see 4.8.2).

## 4. VERIFICATION

4.1 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3)
b. Conformance inspection (see 4.4)
    1. Examination (see 4.5)
    2. Acceptance test (see 4.6)
c. Control test (see 4.7)

4.2 Inspection conditions. Unless otherwise specified (see 6.2), all inspections shall be conducted under the following conditions:

a. Air temperature: 23±10 °C (73±18 °F)
b. Barometric pressure: 96 +6.8/-10.1 kPa (28.5 +2/-3 inches of mercury)
c. Relative humidity: 50±30%

4.3 First article inspection. Unless otherwise specified (see 6.2), first article inspection shall be performed on three fans when a first article sample is required (see 3.1). This inspection shall include the examinations of 4.5 (see table V) and the tests of table III in the order specified in table IV.

4.4 Conformance inspection. Conformance inspection shall include the examinations of 4.5 and the tests of 4.6 and 4.7.

4.5 Examination.

4.5.1 Sampling. Samples from an inspection lot for conformance inspection shall be selected in accordance with ANSI/ASQ Z1.4. Each sample selected shall be inspected as specified herein for the defects listed in table V. Conformance to a lot shall be accepted when zero (0) defects are found and rejected when one (1) or more defects are found.



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TABLE III. Classification of inspections.

| Title | Requirement | Inspection | First article | Quality conformance | | Control |
|---|---|---|---|---|---|---|
| | | | | Examination | Acceptance (100%) | |
| Materials and construction | 3.2, 3.3 and 3.3.2 | 4.8.1 | X | | | |
| Defects (see table V) | 3.3, 3.3.2, 3.5, 3.6 and 3.7 | 4.8.2 | X | X | | |
| Weight | 3.3.1 | 4.8.3 | X | | | |
| Voltage | 3.4.1 | 4.8.4.1 | X | | | X |
| Operational performance | 3.4.2 | 4.8.4.2 | X | | X | X |
| Rated performance | 3.4.3 | 4.8.4.3 | X | | | X |
| Electromagnetic interference suppression | 3.4.4 | 4.8.4.4 | X | | | |
| Insulation resistance | 3.4.5 | 4.8.4.5 | X | | | X |
| Dielectric strength | 3.4.6 | 4.8.4.6 | X | | | X |
| Noise level | 3.4.7 | 4.8.4.7 | X | | | |
| High temperature | 3.4.8.1 | 4.8.4.8.2 | X | | | |
| Low temperature | 3.4.8.2 | 4.8.4.8.3 | X | | | |
| Steam and water- jet cleaning | 3.4.8.3 | 4.8.4.8.4 | X | | | |
| Humidity | 3.4.8.4 | 4.8.4.8.5 | X | | | |
| Shock | 3.4.8.5 | 4.8.4.8.6 | X | | | |
| Vibration | 3.4.8.6 | 4.8.4.8.7 | X | | | X |
| Durability | 3.4.9 | 4.8.4.9 | X | | | |



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TABLE IV. Order of first article inspection. 1/

| Title | Inspection | Sample | | |
|---|---|---|---|---|
| | | 1 | 2 | 3 |
| Defects | 4.8.2 | X | X | X |
| Insulation resistance | 4.8.4.5 | X | X | X |
| Dielectric strength | 4.8.4.6 | X | X | X |
| Rated performance | 4.8.4.3 | X | X | X |
| Weight | 4.8.3 | X | X | X |
| Voltage | 4.8.4.1 | X | X | X |
| Electromagnetic interference suppression | 4.8.4.4 | X | X | X |
| High temperature | 4.8.4.8.2 | X | X | |
| Low temperature | 4.8.4.8.3 | X | X | |
| Steam and waterjet cleaning | 4.8.4.8.4 | X | X | |
| Humidity | 4.8.4.8.5 | X | X | |
| Vibration | 4.8.4.8.7 | X | X | |
| Noise level | 4.8.4.7 | X | X | |
| Shock | 4.8.4.8.6 | X | X | |
| Durability | 4.8.4.9 | | | X |

1/ First article inspection shall include an inspection of contractor records in accordance with 4.8.1.

TABLE V. Classification of defects.

| Category | Defect | Method of examination |
|---|---|---|
| Critical | None | |
| Major | | |
| 101 | Dimensions affecting interchangeability, out of tolerance (see 3.3). | SIE 1/ |
| 102 | Nonconformance in design and construction (see 3.3 and 3.3.2). | Visual |
| 103 | Faulty workmanship affecting performance (see 3.7). | Visual |
| Minor | | |
| 201 | Dimensions not affecting interchangeability, out of tolerance (see 3.3). | SIE |
| 202 | Improper finish (see 3.5). | Visual |
| 203 | Improper identification marking (see 3.6). | Visual |
| 204 | Faulty workmanship affecting appearance (see 3.7). | Visual |

1/ SIE = Standard Inspection Equipment.



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4.6 Acceptance tests (100 percent). Each fan shall be subjected to the test specified in table III. Failure of the test shall be cause for rejection.

4.7 Control tests. Control tests shall be conducted on 1 fan from each lot of 600 units consecutively produced. The fans shall be subjected to the tests specified in table III, in the order specified in table VI.

TABLE VI. Order of control testing.

| Title | Inspection |
|---|---|
| Operational performance | 4.8.4.2 |
| Vibration | 4.8.4.8.7 |
| Insulation resistance | 4.8.4.5 |
| Dielectric strength | 4.8.4.6 |
| Rated performance | 4.8.4.3 |

4.7.1 Failure. If the fan fails to pass any of the control tests, two additional fans may be selected from the lot and subjected to the control tests. Failure of either fan to pass any of the specified control tests shall be cause for rejection.

4.7.2 Disposition of fans. Fans which have been subjected to control testing shall not be delivered on the contract.

4.8 Method of inspection.

4.8.1 Materials and construction. Conformance to 3.2, 3.3 and 3.3.2 shall be determined by inspection of contractor records providing proof or certification that design, construction, processing, and materials conform to requirements. Applicable records shall include drawings, specifications, design data, receiving inspection records, processing and quality control standards, vendor catalogs and certifications, industry standards, test reports, and rating data.

4.8.2 Defects. Conformance to 3.3, 3.3.2, 3.5, 3.6 and 3.7 shall be determined by examination for the defects listed in table V. Examination shall be visual or by measurement with standard inspection equipment.

4.8.3 Weight. To determine conformance to 3.3.1, the fan shall be weighed.

4.8.4 Performance.

4.8.4.1 Voltage. To determine conformance to 3.4.1, attach a variable power supply to the fan and verify that it starts and operates at 11 and 30 V dc.

4.8.4.2 Operational performance. To determine conformance to 3.4.2, the rotor shall be rotated slowly and examined for rubbing and binding. A voltage of 26±2 V dc shall be applied



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to verify that the direction of airflow is as specified, that current does not exceed the specified limit, and that there is no abnormal noise or vibration.

4.8.4.3 Rated performance. To determine conformance to 3.4.3, the fan's airflow rate shall be determined in accordance with AMCA-STD-210.

4.8.4.4 Electromagnetic interference suppression. To determine conformance to 3.4.4, the fan shall be tested in accordance with MIL-STD-461.

4.8.4.5 Insulation resistance. To determine conformance to 3.4.5, the fan insulation resistance shall be tested at 500 V dc potential in accordance with ASTM D257.

4.8.4.6 Dielectric strength. To determine conformance to 3.4.6, the fan insulation dielectric strength shall be tested in accordance with ASTM D149, method A, with the following changes. The testing voltage shall be ramped to 500 V rms at 60 Hertz in one second, and held at 500 V rms for one minute. Evidence of arcing or breakdown shall be considered a failure.

4.8.4.7 Noise level. To determine conformance to 3.4.7, the octave band sound power levels shall be measured in accordance with accepted standard practices for air moving devices.

4.8.4.8 Environmental.

4.8.4.8.1 Test temperature stabilization. Test temperature stabilization is the point at which the test sample will not vary more than 2 °C (3.6 °F) per hour.

4.8.4.8.2 High temperature. To determine conformance to 3.4.8.1, the fan shall be installed in a chamber in a manner that will simulate service usage. The following procedure shall be used. Raise the internal chamber temperature to 71 °C (160 °F) and maintain for a period of 48 hours while insuring the relative humidity is not in excess of 15%. Adjust the internal chamber temperature to 54 °C (130 °F), the highest temperature for which the fan is designed to operate, and maintain until temperature stabilization of the fan is reached (see 4.8.4.8.1). When raising or lowering the temperature of the chamber, the rate of temperature change shall not exceed 10 °C (18 °F) per minute. During the highest operating temperature test phase, the fan shall be subjected to the test specified in 4.8.4.3. Return the fan to standard ambient conditions, stabilize and examine it for evidence of damage or deformation.

4.8.4.8.3 Low temperature. To determine conformance to 3.4.8.2, the fan shall be installed in a chamber in a manner that will simulate service usage. The following procedure shall be used. Lower the internal chamber temperature to -57 °C (-70 °F) and maintain for a period of 24 hours after temperature stabilization. Adjust the internal chamber temperature to -54 °C (-65 °F), the lowest temperature for which the fan is designed to operate, and maintain until temperature stabilization of the fan is reached. When raising or lowering the temperature in the chamber, the rate of temperature change shall not exceed 10 °C (18 °F) per minute. During the lowest operating temperature test phase, the fan shall be subjected to the test specified in



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4.8.4.3. Return the fan to standard ambient conditions, stabilize and examine it for evidence of deformation or damage.

4.8.4.8.4 Steam and waterjet cleaning. To determine conformance to 3.4.8.3, the fan shall be subjected to a cleaning as specified therein. Within 1 hour after completing the steam and waterjet cleaning, the fan shall be subjected to the test specified in 4.8.4.3. The fan shall be operated for 30 minutes and demonstrate no impairment of performance. The fan shall be examined for evidence of damage or deformation.

4.8.4.8.5 Humidity. To determine conformance to 3.4.8.4, the fan shall be tested according to the following procedure:

a. The fan shall be installed as in use in a chamber trap-vented to the atmosphere to prevent the build up of total pressure. No condensation shall be permitted to drip on the fan. Relative humidity shall be determined from dry bulb-wet bulb thermometer comparison method or an equivalent method approved by the procuring activity. Air velocity flowing across the wet bulb shall be not less than 274 meters per minute (m/min) (900 feet per minute (ft/min)). Provision shall be made for controlling the flow of air throughout the internal chamber test space where the velocity of air shall not exceed 46 m/min (150 ft/min). Steam or distilled, demineralized, or deionized water having a hydrogen ion concentration (ph) between 6.0 and 7.2 at 23 °C (73 °F) shall be used to obtain the specified humidity. No rust or corrosive contaminants shall be imposed on the fan by the test facility.
b. After having been dried for 24 hours at 54±2 °C (129±3 °F), the fan shall be conditioned at 23±2 °C (73±3 °F) and 50±10% relative humidity for 24 hours. Raise the internal chamber temperature to 30±2 °C (86±3 °F) and relative humidity to 94±4%. Expose the fan to five continuous 48-hour cycles in accordance with figure 2. After completion of cycling, condition the fan for 24 hours at 23±2 °C (73±3 °F) and 50±10% relative humidity. Accumulated moisture may be removed by turning the fan upside down or shaking. Wiping is not permitted. The fan shall be subjected to the tests specified in 4.8.4.3, 4.8.4.5, and 4.8.4.6 and examined for evidence of damage or deformation.

4.8.4.8.6 Shock. To determine conformance to 3.4.8.5, the fan shall be mounted by normal mounting provisions to a shock machine capable of producing the shock pulse shown in figure 3. Three sawtooth shocks of 40 g and time duration of 11 ms (see figure 3) in each direction shall be applied along three mutually perpendicular axes of the fan (total of 18 shocks). After the shock test, the fan shall be subjected to the test specified in 4.8.4.2 and examined for evidence of damage or deformation.

4.8.4.8.7 Vibration. To determine conformance to 3.4.8.6, the fan shall be attached by its normal mounting means either directly to the vibration exciter or transition table, or by means of a rigid fixture capable of transmitting the vibration conditions specified herein. The fan shall be vibrated along each axis in accordance with the test level specified in figure 1. The sweep time shall be 15 minutes for a sweep frequency range of 5 to 500 to 5 Hz. When fan resonance below 5 Hz is expected, the test curves shall be extended to 2 Hz and the sweep time shall be



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18 minutes. The frequency of applied vibration shall be swept over the specified range logarithmically in accordance with figure 4. The specified sweep time is that of an ascending plus a descending sweep shown and is twice the ascending sweep time shown in figure 4 for the specified range. The sinusodial cycling time shall be 30 minutes per axis. During vibration, the fan shall be operated at 27±2 V dc under ambient temperature conditions. After the vibration test, the fan shall be subjected to the tests specified in 4.8.4.3, 4.8.4.5, and 4.8.4.6 and examined for evidence of damage and deformation.

4.8.4.9 Durability. To determine conformance to 3.4.9, the fan shall be subjected to 1000 hours of continuous operation under the conditions specified therein. No maintenance of any kind shall be allowed during the test except replacement of brushes after not less than 500 hours. At the completion of the durability test, the fan shall be subjected to the tests specified in 4.8.4.3, 4.8.4.5, and 4.8.4.6.

## 5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of materiel is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Department of Defense Agency, or within the Military Department's System Command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

## 6. NOTES

(This section contains information of a general or explanatory nature which may be helpful, but is not mandatory.)

6.1 Intended use. The fan is intended for use on military tracked vehicles in the M2/M3 Bradley Fighting Vehicle Systems (BFVS).

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. If required, the specific issue of individual documents referenced (see 2.2.1, 2.2.2, and 2.3).
c. If first article inspection is not required (see 3.1).
d. Inspection conditions, if other than as specified (see 4.2).
e. First article sample size, if other than as specified (see 4.3).
f. Selection of applicable level and packaging requirements (see 5.1).



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6.3 Definitions.

6.3.1 Recovered materials. "Recovered materials" means materials that have been collected or recovered from solid waste (see 6.4.1.1).

6.3.1.1 Solid waste. "Solid waste" means (a) any garbage, refuse, or sludge from a waste treatment plant, water supply treatment plant, or air pollution control facility; and (b) other discarded material, including solid, liquid, semisolid, or contained gaseous material resulting from industrial, commercial, mining, and agricultural operations, and from community activities. It does not include solid or dissolved material in domestic sewage, or solid or dissolved material in irrigation return flows or industrial discharges which are point sources subject to permits under section 402 of the Clean Water Act, (33 U.S.C. 1342 et seq.), or source, special nuclear, or byproduct material as defined by the Atomic Energy Act of 1954 (42 U.S.C. 2011 et seq.) (Source: Federal Acquisition Regulations, section 23.402).

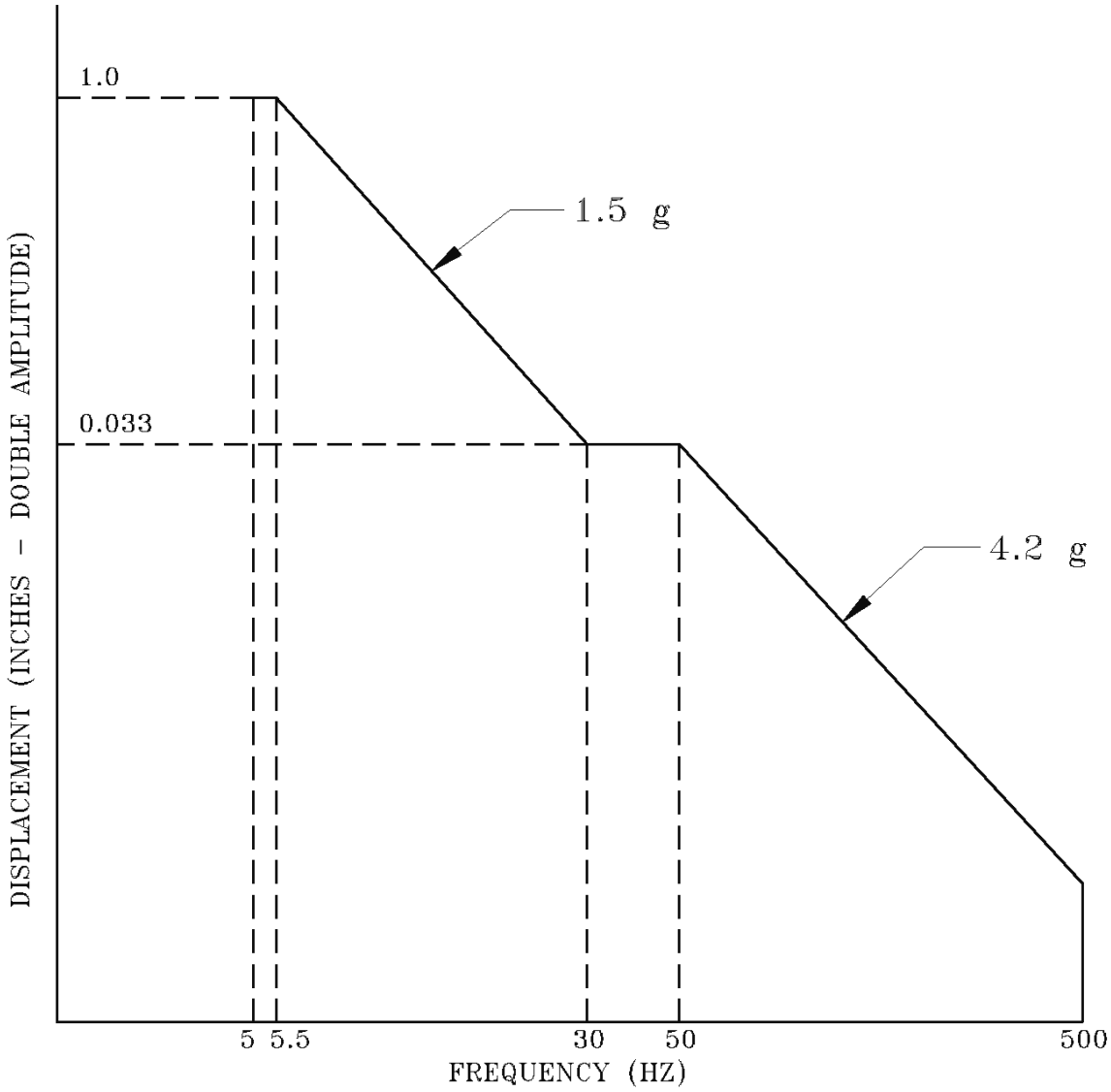
6.4 Subject term (key word) listing.

Electric-powered
Single-stage
Vaneaxial

6.5 Changes from previous issue. The margins of this specification are marked with vertical lines to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.



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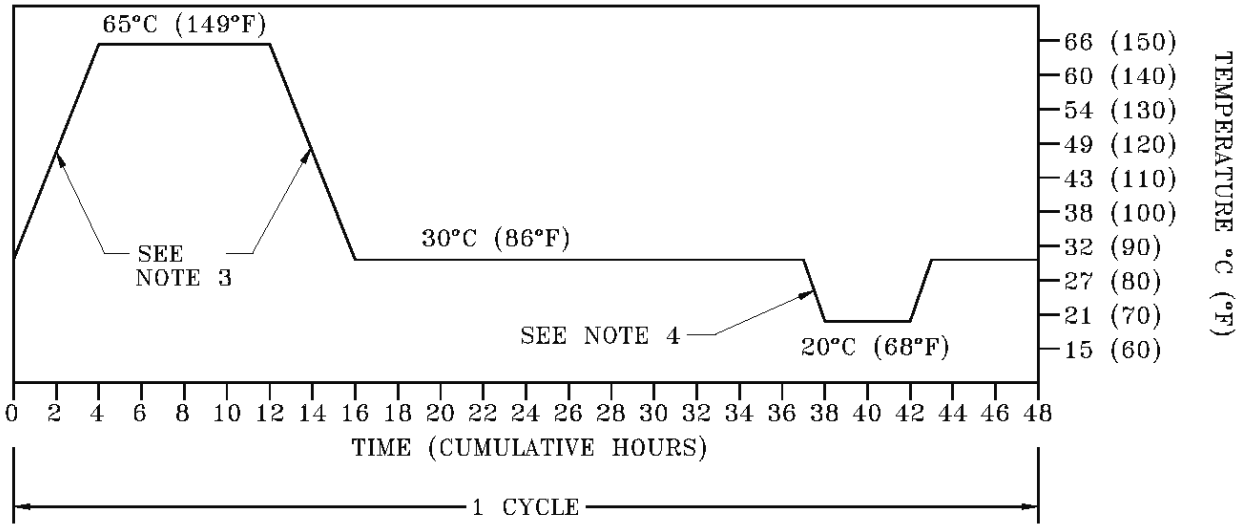

NOTES:

1. Acceleration levels: ±g (peak).
2. The curve shall be extended to 2 Hz when test item resonances below 5 Hz are expected.

FIGURE 1. Vibration test curve.



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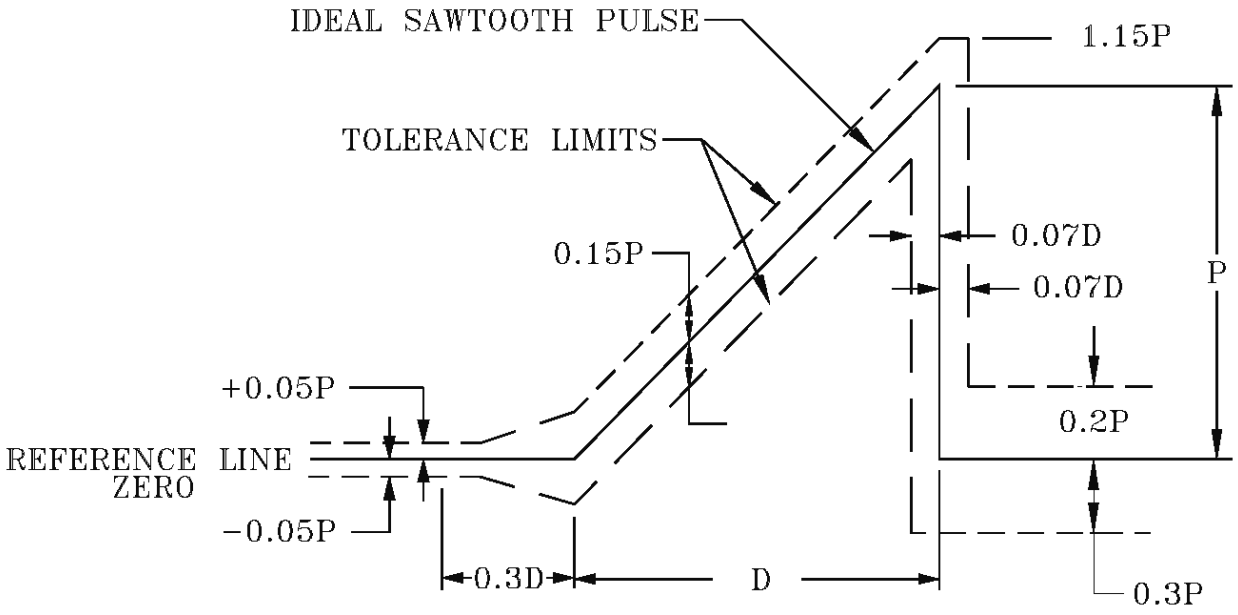

NOTES:

1. Tolerance during temperature change shall be not greater than 3 °C (5 °F).
2. Relative humidity shall be maintained at 94±4% at all times, except that during the descending temperature period, the relative humidity may be permitted to drop as low as 85%.
3. Rate of temperature change between 30 and 65 °C (86 and 149 °F) shall be not less than 8 °C (14.4 °F) per hour.
4. The temperature change in this portion of the curve shall be not less than 10 °C (18 °F).

FIGURE 2. Humidity cycle.



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P = Peak value = 40 g
D = Nominal duration = 11 ms

NOTE: The oscillogram shall include a time about 3D long with a pulse located approximately in the center. The peak acceleration magnitude of the sawtooth pulse is P and its duration is D. The measured acceleration pulse shall be contained between the broken line boundaries. The measured velocity change (which may be obtained by integration of the acceleration pulse) shall be within the limits of Vi±0.1Vi where Vi is the velocity-change associated with the ideal pulse which equals 0.5 PD. The integration to determine velocity change shall extend from 0.4D before the pulse to 0.1D after the pulse.

FIGURE 3. Terminal-peak sawtooth shock pulse configuration and tolerance limits.



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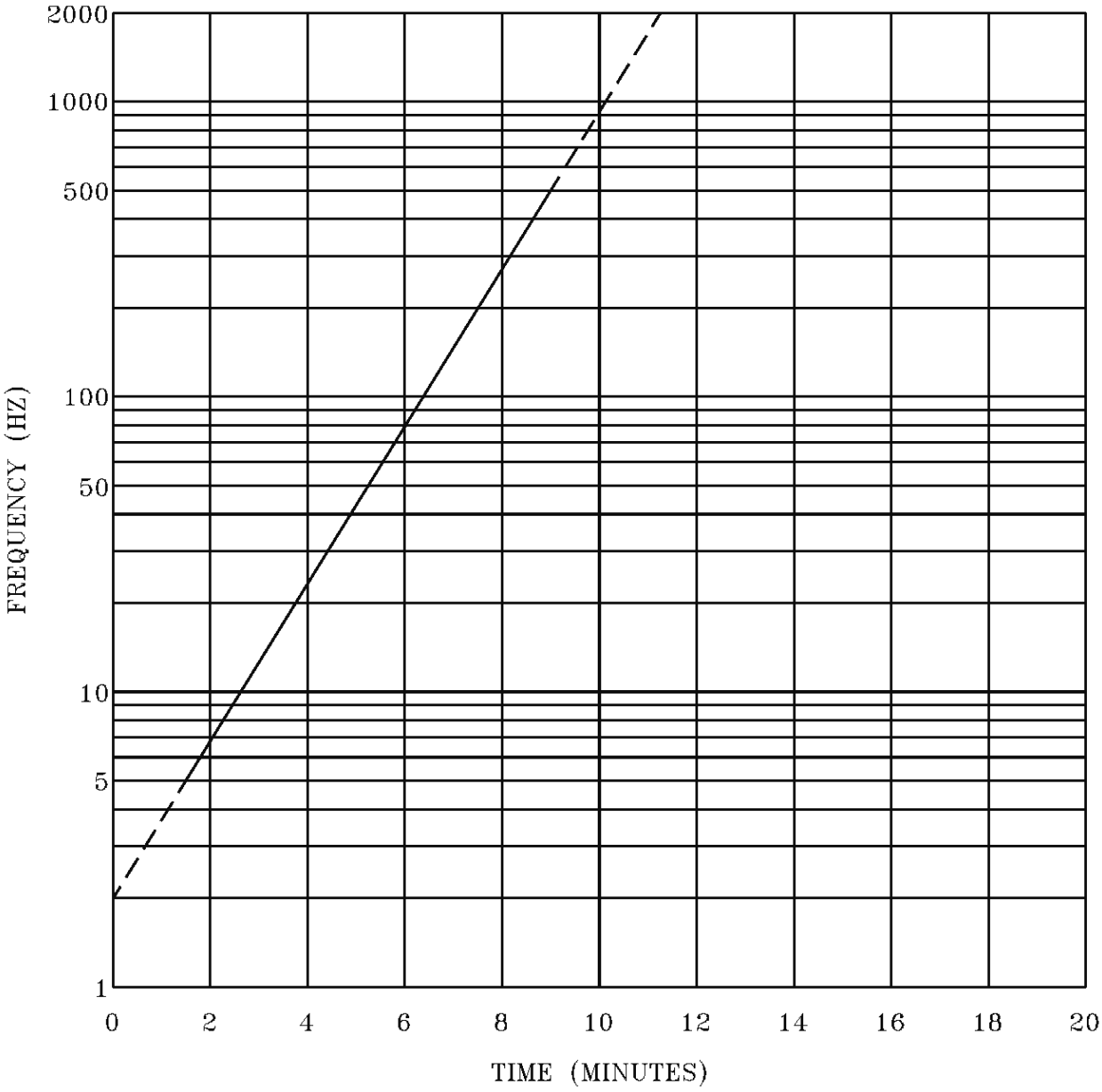



FIGURE 4. Logarithmic sweep.



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